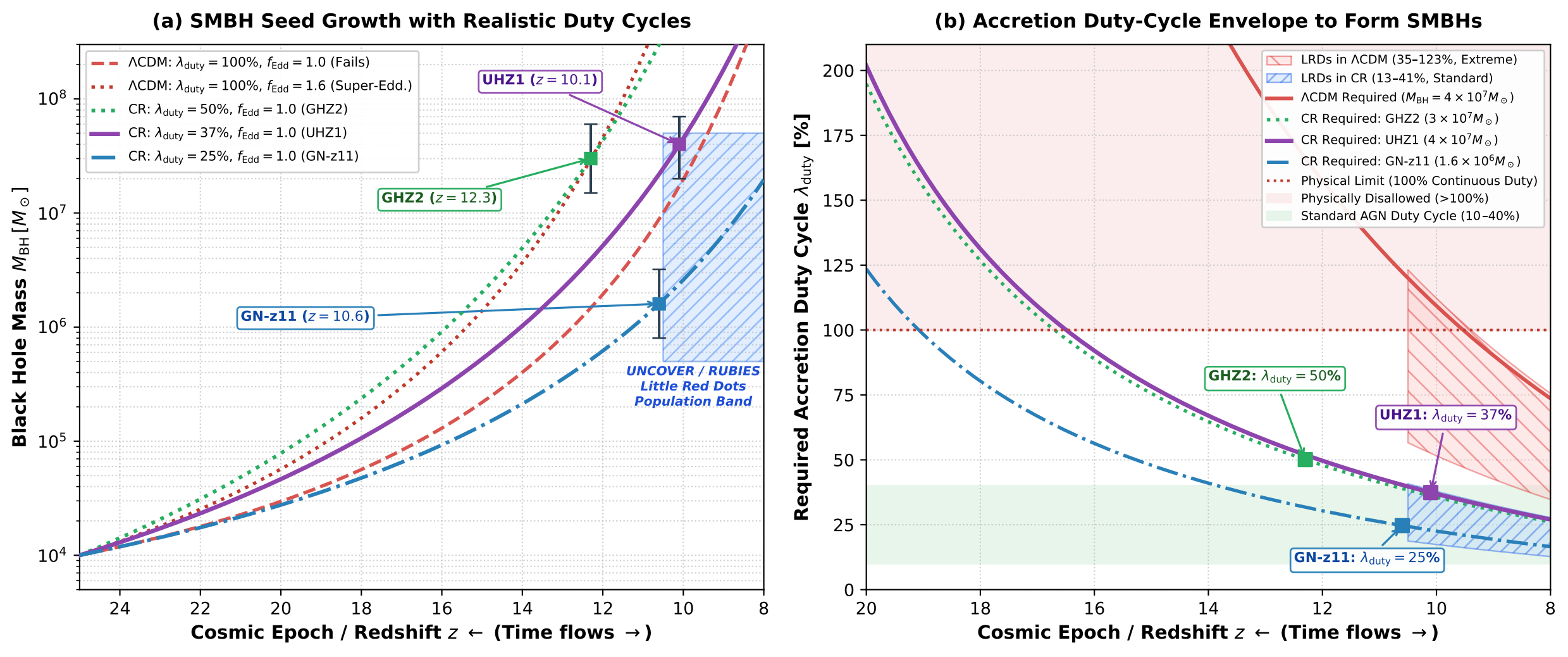

(a) SMBH Seed Growth with Realistic Duty Cycles
ΛCDM: $\lambda_{duty} = 100\%$, $f_{Edd} = 1.0$ (Fails)
ΛCDM: $\lambda_{duty} = 100\%$, $f_{Edd} = 1.6$ (Super-Edd.)
CR: $\lambda_{duty} = 50\%$, $f_{Edd} = 1.0$ (GHZ2)
CR: $\lambda_{duty} = 37\%$, $f_{Edd} = 1.0$ (UHZ1)
CR: $\lambda_{duty} = 25\%$, $f_{Edd} = 1.0$ (GN-z11)
UHZ1 ($z = 10.1$)
GHZ2 ($z = 12.3$)
GN-z11 ($z = 10.6$)
UNCOVER / RUBIES
Little Red Dots
Population Band
Black Hole Mass M_{BH} [$M_\odot$]
Cosmic Epoch / Redshift z ← (Time flows →)
(b) Accretion Duty-Cycle Envelope to Form SMBHs
LRDs in ΛCDM (35–123%, Extreme)
LRDs in CR (13–41%, Standard)
ΛCDM Required ($M_{BH} = 4 \times 10^7 M_\odot$)
CR Required: GHZ2 ($3 \times 10^7 M_\odot$)
CR Required: UHZ1 ($4 \times 10^7 M_\odot$)
CR Required: GN-z11 ($1.6 \times 10^6 M_\odot$)
Physical Limit (100% Continuous Duty)
Physically Disallowed (>100%)
Standard AGN Duty Cycle (10–40%)
GHZ2: $\lambda_{duty} = 50\%$
UHZ1: $\lambda_{duty} = 37\%$
GN-z11: $\lambda_{duty} = 25\%$
Required Accretion Duty Cycle λ_{duty} [%]
Cosmic Epoch / Redshift z ← (Time flows →)